

Pandas Lab Program — Q37 & Q38

Question 37: Create a DataFrame from a Dictionary and Display It

Aim

To write a Pandas program to create a DataFrame from a specified dictionary data and display it.

Source Code

```
import pandas as pd

data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily'],
    'score': [12.5, 9, 16.5, 20, 14],
    'attempts': [1, 3, 2, 3, 2],
    'qualify': ['yes', 'no', 'yes', 'yes', 'no']
}

df = pd.DataFrame(data)
print("DataFrame created from dictionary:\n")
print(df)
```

Output

```
DataFrame created from dictionary:

   name  score  attempts  qualify
0  Anastasia   12.5         1     yes
1     Dima     9.0         3     no
2  Katherine   16.5         2     yes
3     James   20.0         3     yes
4     Emily   14.0         2     no
```

Result

The Python program was executed successfully and a Pandas DataFrame was created from the given dictionary, containing columns 'name', 'score', 'attempts' and 'qualify', and displayed with the default integer index.

Question 38: Create a DataFrame with Custom Index Labels

Aim

To write a Pandas program to create and display a DataFrame from a specified dictionary data which has custom index labels.

Source Code

```
import pandas as pd

data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily'],
    'score': [12.5, 9, 16.5, 20, 14],
    'attempts': [1, 3, 2, 3, 2],
    'qualify': ['yes', 'no', 'yes', 'yes', 'no']
}
labels = ['a', 'b', 'c', 'd', 'e']

df = pd.DataFrame(data, index=labels)
print("DataFrame with custom index labels:\n")
print(df)
```

Output

DataFrame with custom index labels:

	name	score	attempts	qualify
a	Anastasia	12.5	1	yes
b	Dima	9.0	3	no
c	Katherine	16.5	2	yes
d	James	20.0	3	yes
e	Emily	14.0	2	no

Result

The Python program was executed successfully and a Pandas DataFrame was created with the same dictionary data, but with the default index replaced by custom labels ('a', 'b', 'c', 'd', 'e').